

Fast Optimal Design of Micromagnetic Devices Using FastMag and Distributed Evolutionary Algorithm

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A new software framework is presented for fast optimal designs of micromagnetic devices. The developed framework adopts a fast micromagnetic (FastMag) simulator as a micromagnetic solver, and utilizes distributed evolutionary algorithm in Python (DEAP). The framework makes use of a simple Linux utility for resource management (SLURM) in order to run and manage multiple micromagnetic simulations concurrently on a Linux cluster. Various design variables can be designated including material and geometry parameters. The whole design cycle is automated including the mesh generation, and the status of the evolution and multiple simulations can be monitored and managed easily. By combining high computing speed of FastMag and distributed optimization capability of DEAP and SLURM, the optimal designs of micromagnetic devices can be performed rapidly and efficiently. As numerical examples, the optimal designs of a spin-torque oscillator and a magnetic random access memory device are performed.

Index Terms—Distributed optimization, magnetic random access memory (MRAM), micromagnetics, spin-torque oscillator (STO).

I. INTRODUCTION

MICROMAGNETIC devices, such as spin-torque oscillators (STOs) and tunnel magnetoresistance structures, are important components for the advancement of many practical applications, including magnetic recording, magnetic random access memory (MRAM), magnetic sensors, microwave generators, and wireless communications [1]–[4]. In particular, their high compactness, frequency tunability, and good compatibility with the standard complementary metal–oxide–semiconductor process make them ideal for future mobile and power efficient applications [4], [5]. There has been active research work in understanding and characterizing these devices. In many cases, the computations focus on characterization rather than automated multi-parameter optimal design [6]–[10]. Often, on the other hand, there is a need to design a device with a set of operation parameters given a range of geometrical and material parameters. One possibility to achieve this goal is to perform an exhaustive set of simulations over many or all possible parameter combinations. However, such an approach may result in a very large computational time. Optimization techniques can allow performing the device design much more efficiently.

Recently, fast micromagnetic (FastMag) simulators have emerged, including those using finite-difference [11]–[15] and finite-element approaches [16]–[20]. These simulators come in open source or commercial [11]–[21] packages. Many efforts have been made to speed up computations by parallelization [17], [22], [23]. The performance can be further enhanced by implementations on massively parallel graph-

ics processing unit (GPU) computing architectures [12]–[14], [18], [19], [24]–[26]. With a significant speed up, it is now feasible to consider the optimal designs of micromagnetic devices based on optimization algorithms.

In this paper, a new framework is presented for the fast optimal designs of micromagnetic devices. FastMag micromagnetic simulator [19] is adopted as a micromagnetic solver. A distributed evolutionary algorithm in Python (DEAP) [27] is utilized in the optimization module. The presented framework makes use of a simple Linux utility for resource management (SLURM) integrated with the Python and FastMag framework in order to run and manage multiple simulation instances in the parallel optimization process simultaneously on a Linux cluster. Various design variables are pre-defined in a Python library, including material and geometry parameters as well as system meshing. This generality allows readily choosing the design variables in the optimal design. The high performance of FastMag coupled with distributed optimization capability of DEAP and SLURM results in a fast and powerful framework for the optimal designs of micromagnetic devices. As numerical examples, the optimal designs of an STO and a MRAM device are investigated. FastMag as a micromagnetic solver and SLURM as a job submission utility are used as a test bed, which can be extended to other micromagnetic solvers and other utilities.

II. MICROMAGNETIC SOLVER

Micromagnetic phenomena are described by the Landau–Lifshitz–Gilbert (LLG) equation, which assumes that the magnetization is a temporally and spatially continuous function. The generalized LLG equation with the spin-transfer torque (STT) term can be written as [19]

$$\frac{\partial \hat{\mathbf{m}}}{\partial t} = -\gamma \hat{\mathbf{m}} \times \mathbf{H}_{\text{eff}} + \alpha \hat{\mathbf{m}} \times \frac{d\hat{\mathbf{m}}}{dt} + \mathbf{N}_{\text{st}} \quad (1)$$

where $\hat{\mathbf{m}}$ is the normalized magnetization vector, γ is the gyromagnetic ratio, α is the damping constant, and $\mathbf{N}_{\text{st}}$ is

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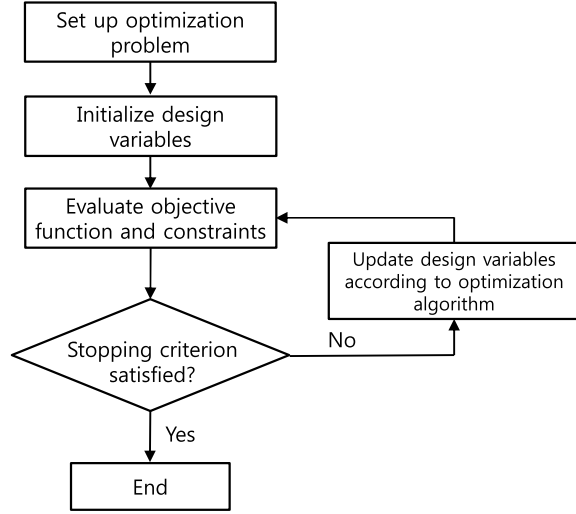


Fig. 1. General flowchart of the optimal design process.

the STT torque. The effective magnetic field $\mathbf{H}_{\text{eff}}$ is given by $\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{ani}} + \mathbf{H}_{\text{exc}} + \mathbf{H}_{\text{ms}}$, where $\mathbf{H}_{\text{ext}}$ is a given external field, $\mathbf{H}_{\text{ani}}$ is the anisotropy field, $\mathbf{H}_{\text{exc}}$ is the exchange field, and $\mathbf{H}_{\text{ms}}$ is the magnetostatic field.

In FastMag, the exchange field $\mathbf{H}_{\text{exc}}$ is calculated as in conventional finite-element method, whereas the magnetostatic field $\mathbf{H}_{\text{ms}}$ is computed directly via the spatial superposition of volume and surface integrals, similar to electromagnetic integral equation methods [28]. The superposition integrals are computed via the adaptive integral (or non-uniform Fourier transform) method for fast and accurate evaluation [19], [25]. FastMag is also equipped with a module for the calculation of the STT effects and non-linear eddy current effects. In addition, FastMag has efficient implicit time evolvers, which make it well suited for simulating complex magnetic devices. FastMag is implemented using C++ and CUDA. It can run both on single- and multi-core CPU computing systems as well as GPU systems.

III. OPTIMIZATION MODULE

The optimization module is implemented with the Python programming language. In the module, the optimization problem is set up with objective function, constraints, and design variables. The objective function is a scalar function that represents the design goal. In addition, various constraints must be satisfied to achieve feasible design. The specified design variables are allowed to change during the design process to improve the objective function within the feasible region. The strength of the proposed optimization module is that it allows implementation and customization of various evolutionary optimization algorithms on distributed computing environment. After the optimization problem is set up, the iterative design process is carried out until the pre-specified stopping criterion is satisfied. Fig. 1 shows the general flowchart of the optimal design process.

A. Setting Up the Optimization Problem

Micromagnetic devices require different sets of physical quantities and parameters in order to set up the optimal design problem than their macroscopic scale counterparts do.

TABLE I
LIST OF DESIGN VARIABLES AVAILABLE FOR
FastMag OPTIMIZATION MODULE

Symbol	Description	Unit	Type
$K_{1(n)}$	first order anisotropy constant in n th material	J/m ³	material
$K_{2(n)}$	second order anisotropy constant in n th material	J/m ³	material
$M_{s(n)}$	saturation magnetization in n th material	T	material
$A_{\text{ex}(n)}$	exchange constant in n th material	J/m	material
$\alpha_{(n)}$	Gilbert damping constant in n th material	-	material
$\theta_{(n)}, \phi_{(n)}$	angle of anisotropy axis in n th material	rad	material
$J_{\text{ex}(n)}$	interlayer exchange between n th and $(n+1)$ th material	J/m ²	material
$d_{(n)}$	interlayer distance between n th and $(n+1)$ th material	nm	geometry
$t_{(n)}$	thickness of the n th layer	nm	geometry
$S_{\text{pp}(n)}$	spin polarization factor in n th material	-	STT ¹
$E_{\text{ff}(n)}$	STT efficiency factor between n th and $(n+1)$ th material	-	STT
$MR_{(n)}$	magnetoresistance factor between n th and $(n+1)$ th material	-	STT
$\beta_{(n)}$	β parameter between n th and $(n+1)$ th material ²	-	STT
$q^{+}_{(n)}$	q^{+} parameter between n th and $(n+1)$ th material ²	-	STT
$q^{-}_{(n)}$	q^{-} parameter between n th and $(n+1)$ th material ²	-	STT
$A_{(n)}$	A parameter between n th and $(n+1)$ th material ²	-	STT
$B_{(n)}$	B parameter between n th and $(n+1)$ th material ²	-	STT
J	applied current	A/m ²	STT
B_x, B_y, B_z	x, y, z components of applied field	T	STT

¹These parameters are for STT (spin-transfer torque) module only.

²These parameters are described in Xiao et al. [30].

For the definition of objective function and constraints, physical quantities, such as precession frequency, precession angle, linewidth, and spectral density of precession, can be used.

Penalty functions are used to handle problem specific constraints. The penalty function gives a disadvantage to objective function value falling in a given region based on the amount of constraint violation. The assigned penalty can be constant or increasing (decreasing for maximization) as the distance from the current design point to a valid solution region increases [29].

As for the design variables, material parameters, such as anisotropy constant, saturation magnetization, exchange constant, and damping constant, can be adopted as well as geometry parameters, such as thickness or the lateral size and shape of each material layer. The design variable library is currently coded for the FastMag solver, so that any combination of the various material and geometry parameters of the micromagnetic device can be set as the design variable vector. It can also be modified to include other micromagnetic solvers, such as Object Oriented MicroMagnetic Framework (OOMMF) [11]. Table I shows the list of material and geometry parameters that can be adopted in the FastMag optimization module. In Table I, the $\beta_{(n)}$ factor is related to the strength of the transverse spin current-induced STT contributions. Typical values for $\beta_{(n)}$ are 0–0.2. The quantities $q^{+}_{(n)}$, $q^{-}_{(n)}$, $A_{(n)}$, and $B_{(n)}$ are device-dependent parameters that define the angular-dependent STT efficiency $\eta(\theta_{12})$ as follows [30]:

$$\eta(\theta_{12}) = \frac{q^{+}_{(n)}}{A_{(n)} + B_{(n)} \cos \theta_{12}} + \frac{q^{-}_{(n)}}{A_{(n)} - B_{(n)} \cos \theta_{12}} \quad (2)$$

where $\cos \theta_{12} = \hat{\mathbf{m}}_{(n)} \cdot \hat{\mathbf{m}}_{(n+1)}$. $\hat{\mathbf{m}}_{(n)}$ and $\hat{\mathbf{m}}_{(n+1)}$ correspond to the magnetizations on the adjacent surface nodes of the faces of two layers (n and $n+1$).

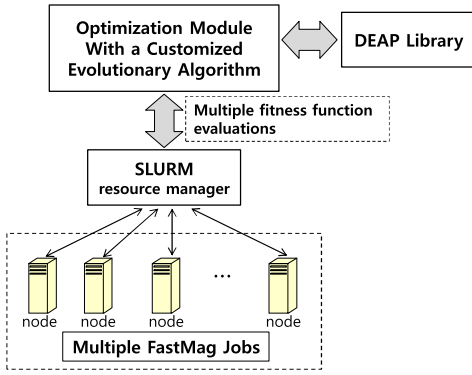


Fig. 2. Block diagram of the proposed optimal design framework.

B. Customization of Distributed Evolutionary Algorithm

For the optimization algorithm to be used with FastMag, the evolutionary algorithm was chosen over the gradient-based optimization method based on the following advantages in the design of micromagnetic devices.

- 1) Micromagnetic devices are dynamic systems with high non-linearities, which make the analytic computation of the gradient of the goal function very difficult and often practically impossible.
- 2) The design space is highly non-uniform, so that the gradient-based methods can fall into local minima. Evolutionary algorithms, on the other hand, search for the global optimum.
- 3) For the evolutionary algorithms, the parallel computing can be achieved, enabling considerable speed up of the whole design process.

The evolutionary algorithm of the optimization module utilizes tools from the DEAP library [27]. DEAP was chosen because it facilitates the customization of sophisticated evolutionary algorithms unlike other black-box software models. It provides basic evolutionary algorithms, such as genetic algorithm (GA), genetic programming, evolution strategy (ES), and particle swarm optimization (PSO). However, it also provides users with ways of creating appropriate types of fitness function (objective function) and individuals (candidate design variables), and customizing and fine-tuning operators, such as mutation, crossover, and selection. Furthermore, it is possible for the users to build their own optimization algorithm for their specific problems with available initializers and operators in the toolbox library.

DEAP is also equipped with tools to distribute computations on multiple processors. However, in the optimization module of FastMag framework, SLURM resource manager was used instead in order to run and manage multiple simulation instances on a Linux cluster during the parallel optimization process. The advantage of using SLURM for distributed computing is that it allows for flexible resource management when other jobs are running on a Linux cluster. Thus, Python codes were written for distributing, executing, and monitoring multiple FastMag jobs on a set of allocated nodes utilizing SLURM commands.

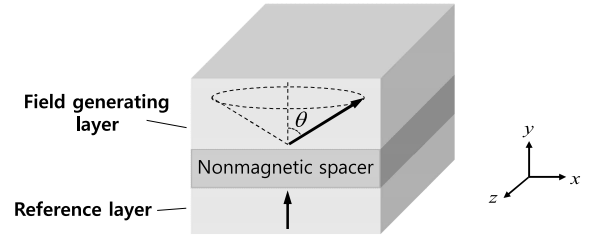


Fig. 3. Schematic illustration of the STO model.

TABLE II
SPECIFICATIONS OF THE INITIAL STO MODELS

Parameter	Value	
$d_{\text{(RL)}}$ (RL thickness) [nm]	7	
$d_{\text{(FGL)}}$ (FGL thickness) [nm]	9	
$d_{\text{(S)}}$ (nonmagnetic spacer thickness) [nm]	3	
Lateral size of the stack [nm ²]	40×40	
$K_{\text{I(FGL)}}$ (FGL anisotropy constant) [J/m ³]	500	
$\alpha_{\text{(FGL)}}$ (FGL damping constant)	0.02	
Parameter	Model 1	Model 2
B_y (y components of applied field) [T]	1.4	1.28
J (applied current) [mA/m ²]	4	3
Analysis Results	Model 1	Model 2
f_{prec} (precession frequency) [GHz]	39.5019	31.4948
θ (precession angle) [deg]	75.7044	61.6829
Lw (linewidth) [GHz]	0.4640	1.6620

Fig. 2 shows a block diagram of the proposed optimal design framework. DEAP library provides tools for customized evolutionary algorithm, whereas SLURM is used for the parallel computing of fitness functions of multiple individuals. The combination of customized evolutionary algorithm and distributed computing, along with powerful FastMag solver, allows for fast and efficient optimal design of micromagnetic devices.

IV. NUMERICAL EXAMPLES

A. Spin-Torque Oscillator Optimization Problem

1) *Problem Definition:* To demonstrate the feasibility and the efficiency of the proposed framework, an optimal design of an STO model is performed as the first numerical example. STOs are a class of nanoscopic microwave oscillators with frequency tunability, broad working temperature, and easy integration with standard silicon technology [5], [31]–[35]. In an STO, STT effect from a spin-polarized current exerts a torque on the magnetization vector of a nanomagnet, leading to persistent magnetization precession. Fig. 3 shows the schematic for the STO. The oscillator consists of a hard reference layer (RL) for spin polarization of the injected current, a non-magnetic spacer, and a field generating (softer) layer (FGL) with high saturation moment.

Generally speaking, it may be difficult to maintain high frequency, stable precession in FGL with low applied field and current, and therein lies the design challenge of an STO. In addition, angle θ (shown in Fig. 3), linewidth, and spectral density of precession must fall within the feasible range. Structurally, the sufficient thickness of FGL is needed for high output power, while the overall stack thickness is limited

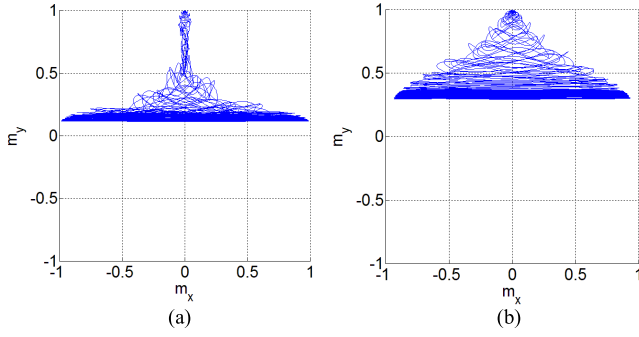


Fig. 4. Trajectories of volume averaged magnetization in FGL (m_x - m_y plane). (a) Model 1 ($B_y = 1.4$ T and $J = 4$ mA/m²). (b) Model 2 ($B_y = 1.28$ T and $J = 3$ mA/m²).

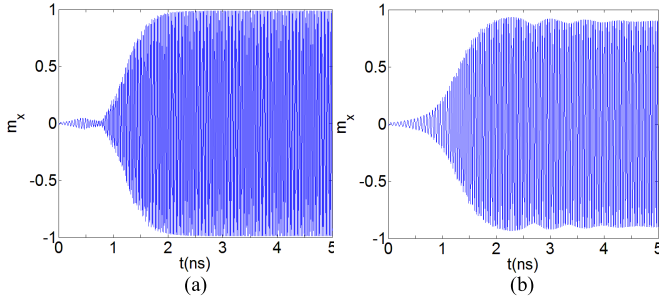


Fig. 5. x -component of magnetization m_x in FGL as a function of time. (a) Model 1 ($B_y = 1.4$ T and $J = 4$ mA/m²). (b) Model 2 ($B_y = 1.28$ T and $J = 3$ mA/m²).

by dimension requirement of the full device, e.g., a write head shield-tip gap. Taking these factors into consideration, the dynamic simulations of two STO models are performed for the time range between 0 and 5 ns using FastMag, and the results are analyzed in order to set up the target precession frequency and other parameters. FastMag uses adaptive time stepping via the backward differentiation formula, and the maximum time step size was unlimited. The resulting average time step size is about 0.1 ps for the time stepping tolerance of $1e-5$. The average mesh size for the model is about 2 nm. Model 1 has higher applied field and the current of 1.4 T and 4 mA/m² compared with model 2 (1.28 T and 3 mA/m²), and other parameters are identical. The specifications and analysis results of the initial STO models are shown in Table II.

The trajectories of the volume averaged magnetization m in FGL are shown in Fig. 4. Figs. 5 and 6 show the x -component of magnetization m_x in FGL as a function of time and the normalized spectral density of m_x precession, respectively.

It can be seen from the analysis that the precession becomes more unstable and the frequency is decreased in model 2, as the applied field and current become lower. In view of the above results, the optimization problem is set up so as to obtain the precession performance comparable with model 1 with applied field and current of model 2, by adjusting other design variables, such as saturation magnetization and exchange constant in FGL. Thus, model 2 is chosen as the initial model, and the target precession frequency is

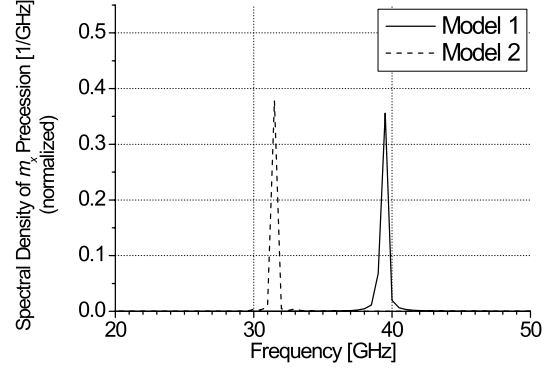


Fig. 6. Normalized spectral density of the m_x precession of the initial STO model 1 ($B_y = 1.4$ T and $J = 4$ mA/m²) and model 2 ($B_y = 1.28$ T and $J = 3$ mA/m²).

set as $f_{\text{target}} = 39$ GHz. The optimization problem is defined as

$$\begin{aligned} \text{Min } F &= c_0(f_{\text{prec}} - f_{\text{target}})^2 \\ \text{s.t. } 30^\circ &\leq \theta \leq 100^\circ, \quad S \geq 0.18 \text{ [1/GHz]}, \quad Lw \leq 0.8 \text{ [GHz]} \end{aligned} \quad (3)$$

where F is the fitness function (objective function), f_{prec} is the calculated precession frequency of the FGL, c_0 is the coefficient for the objective function scaling, S is the normalized spectral density of the precession at the peak frequency, and Lw is the linewidth of the oscillator. Scaling coefficient c_0 can be fine-tuned to improve the performance of some optimization solvers. However, in this example, it is simply set as $c_0 = 1$. The inequality constraints can be imposed by using the quadratic distance penalty functions. For example, the penalty function $h_{\text{angle}}(\theta)$ for precession angle constraint can be written using the quadratic distance to the center of the valid region as follows:

$$h_{\text{angle}}(\theta) = \begin{cases} c_\theta(\theta - 65^\circ)^2 + \Delta_\theta, & \text{if } \theta > 100^\circ \text{ or } \theta < 30^\circ \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where c_θ and Δ_θ are the scaling coefficient and the offset constant for angle constraint penalty function, respectively. These penalty functions are evaluated and added to the objective function F to obtain the total fitness function F_{tot} considering the constraints as follows:

$$F_{\text{tot}} = F + h_{\text{angle}}(\theta) + h_s(S) + h_{lw}(Lw) \quad (5)$$

where $h_s(S)$ and $h_{lw}(Lw)$ are the penalty functions for spectral density and linewidth constraints, respectively.

2) *Optimization Algorithm and Results:* In the numerical example, GA was used as an optimization algorithm. It should be noted that it is easy to implement other standard algorithms such as ES or PSO as well as custom evolutionary algorithm using DEAP library, as was mentioned in Section III. However, standard GA was sufficient to show the feasibility and the efficiency of the overall parallel optimization framework using FastMag, which is the main focus of this paper.

In GA, a proportion of the current population is selected based on the fitness values to breed a new generation. DEAP library provides various genetic operators to choose

TABLE III
OPTIMIZATION AND ANALYSIS PARAMETERS (STO)

Optimization Parameters	Case 1	Case 2	Case 3
Number of design variables	3	4	5
Population size	18	23	27
Common Parameters			
f_{target} [GHz]		39	
cpxb (crossover ratio)		0.85	
alpha (blend ratio)		0.5	
mutpb (mutation ratio)		0.1	
Selection tournament size		2	
B_y (applied field) [T]		1.28	
J (applied current) [mA/m ²]		3	

from, and tournament selection operator was used for this example. Tournament selection involves several tournaments among randomly chosen individuals of fixed number (tournament size), and the winner of each tournament is selected [36].

After the selection process, a new generation of population is created from those selected by two genetic operations, i.e., crossover and mutation. Crossover ratio and mutation ratio determine the probability of each operation happening for random individuals, respectively. For crossover, a blend crossover operator was used that combines variable values using weighted average of two parents. Blend ratio (alpha) adjusts the weighting of each parent [37]. For mutation operator, polynomial mutation as implemented in original NSGA-II algorithm was adopted [38]. For the stopping criterion of GA, the fitness limit value of $1e-7$ was chosen. The algorithm stops when the fitness value of the best individual in the current generation is less than or equal to the fitness limit. Finally, for fitness function, F_{tot} with penalty terms as defined in (3)–(5) was used.

Three optimal design cases are considered with different numbers of design variables to test the performance of the proposed optimization framework: case 1 with three design variables, case 2 with four design variables, and case 3 with five design variables. Applied field and current are the same as those of model 2 in Section IV-A.1) ($B_y = 1.28$ T and $J = 3$ mA/m²) for all cases. Tables III and IV show various parameters of optimization algorithm mentioned earlier, as well as the list design variables and the optimization results, including the total number of FastMag calls. Figs. 7–9 compare the trajectories of the volume averaged magnetization m in FGL, the x -component of magnetization m_x in FGL, and the normalized spectral density of m_x precession between initial model and optimized ones, respectively. The qualities of optimal solutions are great in all cases, i.e., the precession frequency approaches the exact target frequency of 39 GHz, and fitness function values approach 0. Although the precession characteristics of the optimal solutions are nearly identical and much improved from the initial model, as can be seen from Figs. 7–9, and the design variables show noticeable differences between the cases. These results suggest that there are multiple optimal solutions in the specified design space. In addition, the total number of FastMag calls is not directly proportional to the number of design variables. It should be noted that the combinations

TABLE IV
DESIGN VARIABLES AND OPTIMIZATION RESULTS (STO)

Design Variables	Optimized (case 1)	Optimized (case 2)	Optimized (case 3)
$K_{1(\text{RL})}$ [J/m ³]	473,587	475,917	481,545
A_{ex} [J/m]	1.2495e-11	1.4351e-11	1.3594e-11
$M_{\text{s(FGL)}}$ [T]	1.1199	1.1928	1.2024
$\alpha_{(\text{RL})}$	0.2 (fixed)	0.1781	0.0677
$M_{\text{s(RL)}}$ [T]	1.1 (fixed)	1.1 (fixed)	1.0691
Optimization Results			
f_{prec} [GHz]	39.0001	38.9998	39.0001
θ [deg]	97.5916	97.1630	97.2042
L_w [GHz]	0.3800	0.3880	0.3840
F (fitness value)	1.1025e-8	5.2441e-8	4.7610e-9
No. of generations	9	10	7
Total no. of FastMag calls	145	215	177

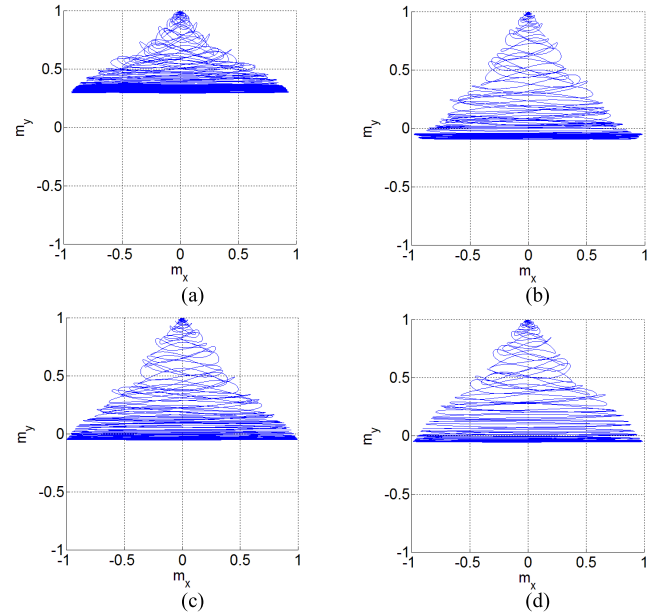


Fig. 7. Trajectories of volume averaged magnetization in FGL (m_x – m_y plane) ($B_y = 1.28$ T and $J = 3$ mA/m² for all models). (a) Initial model. (b) Case 1 (optimized with three design variables). (c) Case 2 (optimized with four design variables). (d) Case 3 (optimized with five design variables).

of ten point sweeps of the five design variables (case 3) would have taken 10^5 FastMag calls, about 565 times more than the optimal design case. These results confirm the feasibility of the proposed optimization framework.

3) *Comparison With Deterministic Optimization*: To test the efficiency and the speed of the proposed optimization framework, the quality of optimal solutions and performance of the optimization module is compared with those obtained by deterministic optimization method. For the deterministic optimizer, fmincon solver in the MATLAB optimization toolbox is used, which is a gradient-based optimizer. The optimization problem for the deterministic method is given by (3), the same as in Section IV-A.2). In fmincon, the gradient of the objective function is approximated by the finite-difference method. Distributed optimization is not adopted for fmincon. Table V shows the comparison of optimization results between the two methods, and Figs. 10 and 11 show the plots of

TABLE V
COMPARISON OF OPTIMIZATION RESULTS

F_{tot} (total fitness value)	fmincon	Proposed
Case 1 (3 design variables)	6.7397e-4	1.1025e-8
Case 2 (4 design variables)	1.4980e-3	5.2441e-8
Case 3 (5 design variables)	4.9244e-5	4.7610e-9
Total no. of FastMag calls	fmincon	Proposed
Case 1	163	145
Case 2	64	215
Case 3	150	177
Total Computation Time	fmincon	Proposed
Case 1	49h 56m 42s	12h 42m 48s
Case 2	21h 08m 56s	10h 33m 50s
Case 3	44h 33m 53s	5h 24m 23s

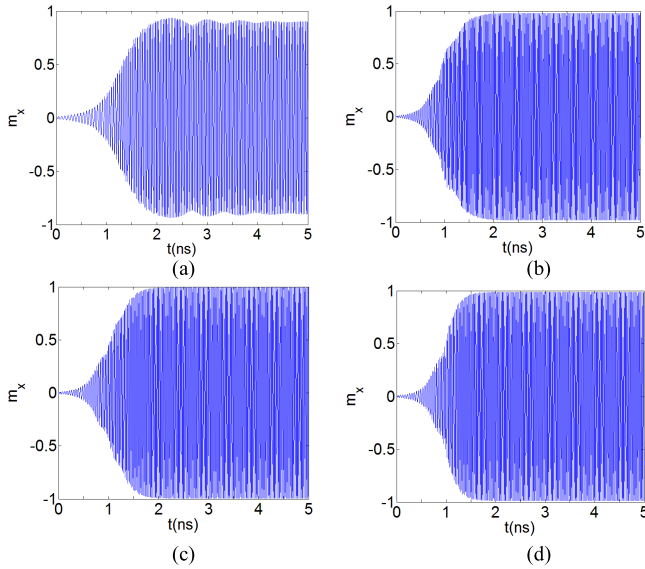


Fig. 8. x -component of magnetization m_x in FGL as a function of time ($B_y = 1.28$ T and $J = 3$ mA/m² for all models). (a) Initial model. (b) Case 1 (optimized with three design variables). (c) Case 2 (optimized with four design variables). (d) Case 3 (optimized with five design variables).

convergence of optimal solutions and total computation time, respectively. It should be noted that each generation of GA and iteration of fmincon in Fig. 10 consists of a number of FastMag calls as required by each optimizer. The final fitness values (the quality of optimal solutions) for the proposed framework were at least 10^4 times better than those for fmincon. In addition, even though the total numbers of FastMag calls are similar in both methods, significant reduction in total computation time can be observed for the proposed framework compared with the fmincon solver because of the full utilization of the distributed optimization.

Implementing distributed optimization for deterministic method (fmincon) is not straightforward, because it is not population-based. Deterministic method comprises of two computationally intensive parts: the calculation of the objective function itself and the estimation of its gradient. The potential acceleration by implementing distributed optimization only affects the gradient estimation part, and the time saving in that part will be proportional to the number of design variables. On the other hand, the maximum acceleration of the proposed framework is proportional to the population size,

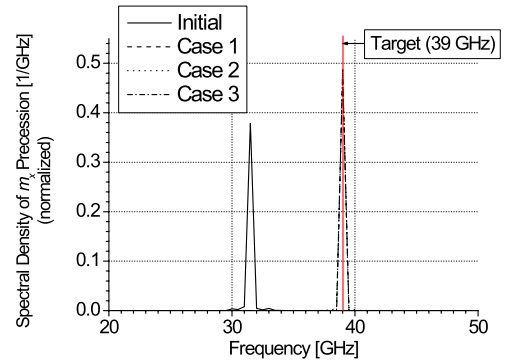


Fig. 9. Normalized spectral density of the precession of the initial and optimized STO models ($B_y = 1.28$ T and $J = 3$ mA/m² for all cases).

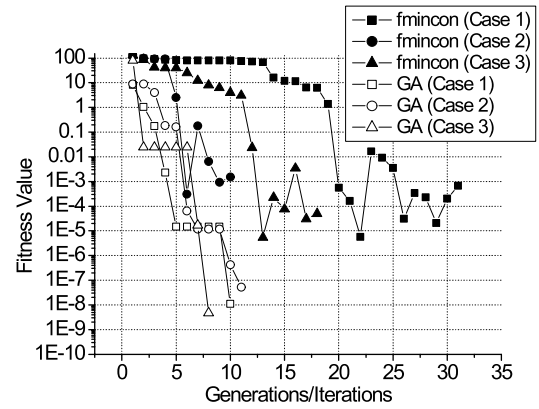


Fig. 10. Comparison of fitness function convergence between fmincon and the proposed framework (STO).

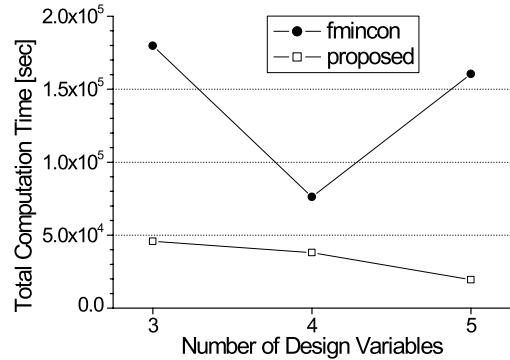


Fig. 11. Comparison of total computation time between fmincon and the proposed framework (STO).

which is much greater than the number of design variables (individual size). Hence, the proposed framework will result in a more efficient use of distributed computing resources even when the deterministic method utilizes distributed optimization. On another note, it should be mentioned that the total computation time can be affected by workload status of other jobs running on the cluster. However, it is still an effective estimate of overall performance of the optimization framework.

It should be noted that for the results in Table V and Figs. 10 and 11, the stopping criteria for fmincon were set as

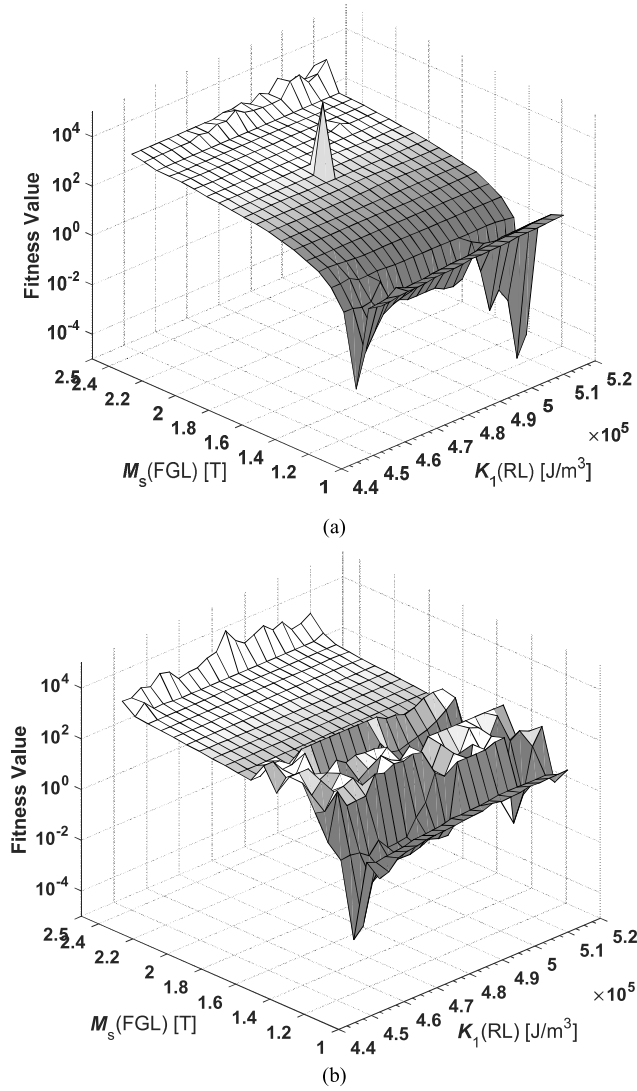


Fig. 12. 3-D surface plots of the STO design space with respect to two design variables $K_1(\text{RL})$ and $M_s(\text{FGL})$ [$\alpha(\text{RL}) = 0.0677$ and $M_s(\text{RL}) = 1.0691$ T for both plots]. (a) When $A_{\text{ex}} = 1.3594\text{e-}11$ J/m. (b) When $A_{\text{ex}} = 1.1\text{e-}11$ J/m.

$\text{TolX} = 1.0\text{e-}5$ and $\text{TolFun} = 1.0\text{e-}5$, where TolX is a lower bound on the change in the norm of the design variable vector, and TolFun is a lower bound on the change in the fitness function value for one step. Decreasing TolX and TolFun for *fmincon* did not result in much improvement of the fitness function. For example, when TolX and TolFun values were decreased to $1.0\text{e-}7$ (from the original values of $1.0\text{e-}5$) for case 3 (five design variables), the total number of FastMag calls was increased from 150 to 224, but the fitness function value was merely decreased from $4.9244\text{e-}5$ to $4.8213\text{e-}5$. For case 2 (four design variables), when TolX and TolFun values were decreased to $1.0\text{e-}7$ from $1.0\text{e-}5$, the total number of FastMag calls was increased from 64 to 75, but the objective function remained at the same value of $1.4980\text{e-}3$. This is due to the fact that the increased number of function calls or smaller tolerance for convergence criteria does not necessarily improve the quality of the objective function very much when the gradient-based optimization method, such as *fmincon*, has

fallen into a local minimum. Consequently, the fitness value quality comparable with that of the proposed method could not be obtained using *fmincon*.

To visualize and further investigate the complexity and non-linearity of the STO design space, 21 point sweeps of the two design variables, $K_1(\text{RL})$ and $M_s(\text{FGL})$, are performed for the total of $21 \times 21 = 441$ FastMag calls. This procedure is repeated for two different values of A_{ex} , namely, $A_{\text{ex}} = 1.3594\text{e-}11$ J/m (from Table IV) and $A_{\text{ex}} = 1.1\text{e-}11$ J/m. Other parameters are fixed at the values taken from the optimal solution (case 3) of the proposed method in Table IV. Fig. 12 shows the 3-D surface plots of the design space with x -, y -, and z -axes corresponding to $K_1(\text{RL})$, $M_s(\text{FGL})$, and fitness value, respectively. Even though the sweep resolution is not high, it can be seen that the design space is highly non-uniform and non-linear with many local minima. Furthermore, the 3-D surface plots show considerably different characteristics for two different values of A_{ex} . The complexity of the STO design space results from the non-linear interfacial STT terms, $\mathbf{N}_{\text{st}}$ in (1), that are added to the classical LLG equation [30] in this model. It is expected that the complexity of the design space will grow even more as the number of design variables increases. These results show that the evolutionary algorithm is better suited for the optimal designs of the micromagnetic devices than the gradient-based optimization method, and the distributed optimization of the proposed framework is a powerful and efficient tool for the fast optimal designs of micromagnetic devices.

B. MRAM Optimization Problem

1) *Problem Definition*: In the past decade, STT-MRAM that utilizes STT writing technology emerged as a promising candidate for a future universal memory with low-power consumption, high endurance, fast speed, and excellent scalability [39]–[41]. One of the important design aspects of STT-MRAM is its thermal stability, since spontaneous magnetization reversal of the storage layer by thermal agitation can lead to a loss of the recorded data [42], [43]. Thermal stability of the magnetic system can be estimated by the energy barrier between the stable states [44]. In this section, an optimal shape design of a perpendicular STT-MRAM is performed using the proposed framework as the second numerical example in order to obtain target energy barriers.

Fig. 13 shows the lateral and top views of the perpendicular STT-MRAM model. The circularly or elliptically shaped stack consists of a free layer (FL) that stores data by switching magnetization direction between two stable states, a MgO insulator junction, RL, and synthetic antiferromagnet layer 1 (SAF1) and SAF2 that are antiferromagnetically exchange coupled. The thermal stability of the FL is affected by the magnetostatic field generated by FL, SAF1, and SAF2, such that the energy barrier can be different for the two (up or down) states of the FL. However, the thicknesses of SAF1 and SAF2 as well as the radii can be found to lead to a particular energy barrier, which is the same for both FL states.

The energy barriers of the MRAM are calculated via nudged elastic band method. This method finds minimum energy

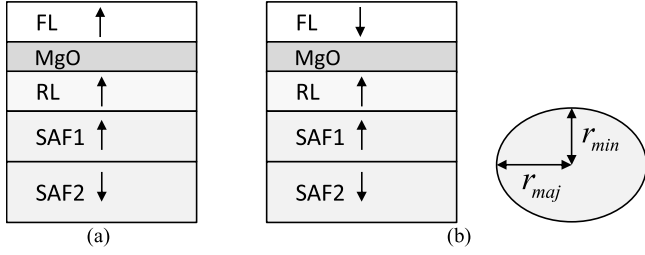


Fig. 13. Schematic of the perpendicular STT-MRAM model. (a) Up state (b) Down state and top view.

TABLE VI
OPTIMIZATION AND ANALYSIS PARAMETERS (MRAM)

Parameters	Value
Number of design variables	4
Population size	30
c_1 (scaling coefficient)	1
c_2 (scaling coefficient)	1
c_3 (scaling coefficient)	0.5
c_4 (scaling coefficient)	0.2
t_{FL} (free layer thickness) [nm]	0.9
t_{RL} (reference layer thickness [nm])	1
t_{MgO} (MgO layer thickness) [nm]	1

path (MEP) between local energy minima in high-dimensional energy space and is a rigorous way to compute the saddle point between the initial and final energy states [44], [45]. For the considered range of parameters, the MEP is via a domain wall propagating along the major axis direction. The design goal of the STT-MRAM is to obtain sufficiently high-energy barriers for both up→down and down→up energy paths while maintaining both energy barriers in a similar range and keeping the area of the stack ellipse small. The optimization problem can be defined as

$$\text{Min: } F = c_1(E_{B1} - E_{B1o})^2 + c_2(E_{B2} - E_{B2o})^2 + c_3(E_{B1} - E_{B2})^2 + c_4 r_{maj} r_{min} \quad (6)$$

where F is the fitness function, E_{B1} , E_{B1o} and E_{B2} , E_{B2o} are the calculated energy barrier and target one for up→down and down→up energy path, respectively, r_{maj} and r_{min} are the major and minor radii of the stack ellipse, and c_1 , c_2 , c_3 , and c_4 are the scaling coefficients. The first three terms of F correspond to the energy barrier differences, while the fourth term ($c_4 r_{maj} r_{min}$) is proportional to the area of the stack ellipse.

2) *Optimization Results:* For the optimization algorithm for MRAM, GA was adopted as in the STO design case. For stopping criterion of GA, the fitness limit value of 240 was chosen. This value corresponds to the case when the first three energy terms of F are zero ($E_{B1} = E_{B2} = E_{B1o} = E_{B2o}$) and the radii product term is $r_{maj} r_{min} = 1200$ (for example, $r_{maj} = 40$ nm and $r_{min} = 30$ nm) with $c_4 = 0.2$. The optimization stops when the fitness value of the best individual in the current generation is less than or equal to the fitness limit. The design variables are the aspect ratio of the ellipse $a_r = r_{maj}/r_{min}$, the minor radius r_{min} , and the thicknesses of the SAF1 and SAF2 layers (t_{SAF1} and t_{SAF2}). Table VI shows various optimization and analysis parameters, and

TABLE VII
DESIGN VARIABLES AND OPTIMIZATION RESULTS (MRAM)

Design Variables	Lower Bound	Upper Bound	Initial	Optimized
a_r	1	1.5	1.25	1.1415
r_{min} [nm]	10	70	40	31.8104
t_{SAF1} [nm]	3	7	5	3.2012
t_{SAF2} [nm]	4	14	9	7.3911
Optimization Results		target	Initial	Optimized
E_{B1} [k_B 300K]		50	99.57	48.68
E_{B2} [k_B 300K]		50	32.52	48.31
F (fitness value)		-	-	235.679
No. of generations		-	-	16
Total no. of FastMag calls		-	-	416
Total Computation Time		-	-	38h 47m 59s

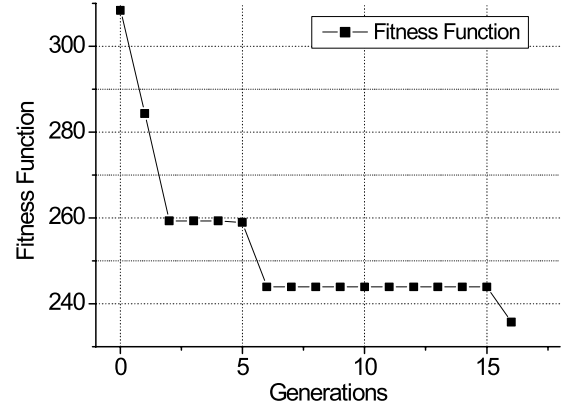


Fig. 14. Convergence of the fitness function (MRAM).

Table VII shows the design variables and the optimization results, including the energy barriers of up→down and down→up paths (E_{B1} and E_{B2}) before and after the optimization. It can be seen that the energy barriers for both paths approach the target value of 50 k_B 300 K very closely. The minor radius r_{min} approaches 31.8 nm and SAF1 thickness settled near the lower bound of 3 nm. Fig. 14 shows the convergence of the fitness function.

V. CONCLUSION

A new optimization framework was developed for the fast optimal design of micromagnetic devices. The proposed framework adopts FastMag as a micromagnetic solver, and utilizes DEAP library and SLURM resource manager for distributed evolutionary optimization. DEAP library facilitates implementation and customization of population-based optimization algorithms, while SLURM enables flexible resource management of computing nodes. The combination of high performance and flexibility of FastMag and GPU computing with distributed optimization capability of DEAP and SLURM resulted in a fast and powerful framework for the design of micromagnetic devices. As numerical examples, the optimal designs of an STO and MRAM were performed to obtain target precession frequency or energy barriers. Comparison of the results with the conventional deterministic optimization method shows the efficiency of the proposed framework. It is expected that the developed framework can be used for the fast

optimal designs of various micromagnetic devices. It can also be extended to include other micromagnetic solvers. Moreover, the additional levels of parallelization can be achieved by running multiple jobs on each GPU device.

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REFERENCES

- [1] X. Zhu and J.-G. Zhu, "Bias-field-free microwave oscillator driven by perpendicularly polarized spin current," *IEEE Trans. Magn.*, vol. 42, no. 10, pp. 2670–2672, Oct. 2006.
- [2] J.-G. Zhu, X. Zhu, and Y. Tang, "Microwave assisted magnetic recording," *IEEE Trans. Magn.*, vol. 44, no. 1, pp. 125–131, Jan. 2008.
- [3] J.-G. Zhu and X. Zhu, "Spin transfer induced noise in CPP read heads," *IEEE Trans. Magn.*, vol. 40, no. 1, pp. 182–188, Jan. 2004.
- [4] H. S. Choi *et al.*, "Spin nano-oscillator-based wireless communication," *Sci. Rep.*, vol. 4, Jun. 2014, Art. no. 5486.
- [5] Z. Zeng, G. Finocchio, and H. Jiang, "Spin transfer nano-oscillators," *Nanoscale*, vol. 5, no. 6, pp. 2219–2231, 2013.
- [6] J. C. Slonczewski, "Current-driven excitation of magnetic multilayers," *J. Magn. Magn. Mater.*, vol. 159, nos. 1–2, pp. L1–L7, Jun. 1996.
- [7] L. Berger, "Emission of spin waves by a magnetic multilayer traversed by a current," *Phys. Rev. B*, vol. 54, pp. 9353–9358, Oct. 1996.
- [8] E. B. Myers, D. C. Ralph, J. A. Katine, R. N. Louie, and R. A. Buhrman, "Current-induced switching of domains in magnetic multilayer devices," *Science*, vol. 285, no. 5429, pp. 867–870, Aug. 1999.
- [9] J. A. Katine, F. J. Albert, R. A. Buhrman, E. B. Myers, and D. C. Ralph, "Current-driven magnetization reversal and spin-wave excitations in Co/Cu/Co pillars," *Phys. Rev. Lett.*, vol. 84, no. 14, pp. 3149–3152, 2000.
- [10] A. Thiaville, Y. Nakatani, J. Miltat, and Y. Suzuki, "Micromagnetic understanding of current-driven domain wall motion in patterned nanowires," *Europhys. Lett.*, vol. 69, no. 6, pp. 990–996, Mar. 2005.
- [11] M. J. Donahue and D. G. Porter, "OOMMF user's guide," Dept. Math. Softw. Group, Nat. Inst. Standards Technol., Gaithersburg, MD, USA, Tech. Rep. NISTIR 6376, Sep. 1999.
- [12] S. Fu, W. Cui, M. Hu, R. Chang, M. J. Donahue, and V. Lomakin, *OOMMF on GPU*, accessed on May 24, 2016. [Online]. Available: <http://cem.ucsd.edu/OOMMF.html>
- [13] A. Vansteenkiste and B. Van de Wiele, "MuMax: A new high-performance micromagnetic simulation tool," *J. Magn. Magn. Mater.*, vol. 323, no. 21, pp. 2585–2591, Nov. 2011.
- [14] L. Lopez-Diaz *et al.*, "Micromagnetic simulations using graphics processing units," *J. Phys. D, Appl. Phys.*, vol. 45, no. 32, p. 323001, Jul. 2012.
- [15] G. Recio and C. Estébanez, "Micromagnetic modeling on magnetisation dynamics with lossy magnetic material in thin film heads by FDTD calculations," *Appl. Comput. Electromagn. Soc. J.*, vol. 27, no. 9, pp. 717–725, Sep. 2012.
- [16] T. Fischbacher, M. Franchin, G. Bordignon, and H. Fangohr, "A systematic approach to multiphysics extensions of finite-element-based micromagnetic simulations: Nmag," *IEEE Trans. Magn.*, vol. 43, no. 6, pp. 2896–2898, Jun. 2007.
- [17] W. Scholz *et al.*, "Scalable parallel micromagnetic solvers for magnetic nanostructures," *Comput. Mater. Sci.*, vol. 28, no. 2, pp. 366–383, Oct. 2003.
- [18] A. Kakay, E. Westphal, and R. Hertel, "Speedup of FEM micromagnetic simulations with graphical processing units," *IEEE Trans. Magn.*, vol. 46, no. 6, pp. 2303–2306, Jun. 2010.
- [19] R. Chang, S. Li, M. V. Lubarda, B. Livshitz, and V. Lomakin, "Fast-Mag: Fast micromagnetic simulator for complex magnetic structures (invited)," *J. Appl. Phys.*, vol. 109, no. 7, p. 07D358, 2011.
- [20] V. Lomakin, *FastMag: Fast Micromagnetic Simulator for Complex Magnetic Structures*, accessed on May 24, 2016. [Online]. Available: <http://cem.ucsd.edu/FastMag.html>
- [21] M. R. Scheinfein, *LLG Micromagnetics Simulator*, accessed on May 24, 2016. [Online]. Available: <http://llgmicro.home.mindspring.com>
- [22] K. Takano, E.-A. Salhi, M. Sakai, and M. Dovek, "Write head analysis by using a parallel micromagnetic FEM," *IEEE Trans. Magn.*, vol. 41, no. 10, pp. 2911–2913, Oct. 2005.
- [23] M. J. Donahue, "Parallelizing a micromagnetic program for use on multiprocessor shared memory computers," *IEEE Trans. Magn.*, vol. 45, no. 10, pp. 3923–3925, Oct. 2009.
- [24] S. Fu, W. Cui, M. Hu, R. Chang, M. J. Donahue, and V. Lomakin, "Finite-difference micromagnetic solvers with the object-oriented micromagnetic framework on graphics processing units," *IEEE Trans. Magn.*, vol. 52, no. 4, Apr. 2016, Art. no. 7100109.
- [25] R. Chang, M. A. Escobar, S. Li, M. V. Lubarda, and V. Lomakin, "Accurate evaluation of exchange fields in finite element micromagnetic solvers," *J. Appl. Phys.*, vol. 111, no. 7, p. 07D129, 2012.
- [26] M. A. Escobar, M. V. Lubarda, S. Li, R. Chang, B. Livshitz, and V. Lomakin, "Advanced micromagnetic analysis of write head dynamics using Fastmag," *IEEE Trans. Magn.*, vol. 48, no. 5, pp. 1731–1737, May 2012.
- [27] F.-A. Fortin, F.-M. De Rainville, M.-A. G. Gardner, M. Parizeau, and C. Gagné, "DEAP: Evolutionary algorithms made easy," *J. Mach. Lang. Res.*, vol. 13, pp. 2171–2175, Jul. 2012.
- [28] J.-M. Jin, V. V. Liepa, and C.-T. Tai, "A volume-surface integral equation for electromagnetic scattering by inhomogeneous cylinders," *J. Electromagn. Waves Appl.*, vol. 2, nos. 5–6, pp. 573–588, 1988.
- [29] C. A. C. Coello, "Theoretical and numerical constraint-handling techniques used with evolutionary algorithms: A survey of the state of the art," *Comput. Methods Appl. Mech. Eng.*, vol. 191, pp. 1245–1287, Jan. 2002.
- [30] J. Xiao, A. Zangwill, and M. D. Stiles, "Macrospin models of spin transfer dynamics," *Phys. Rev. B*, vol. 72, p. 014446, Jul. 2005.
- [31] S. I. Kiselev *et al.*, "Microwave oscillations of a nanomagnet driven by a spin-polarized current," *Nature*, vol. 425, no. 6956, pp. 380–383, Sep. 2003.
- [32] W. H. Rippard, M. R. Pufall, S. Kaka, S. E. Russek, and T. J. Silva, "Direct-current induced dynamics in Co₉₀Fe₁₀/Ni₈₀Fe₂₀ point contacts," *Phys. Rev. Lett.*, vol. 92, no. 2, p. 027201, Jan. 2004.
- [33] I. N. Krivorotov, N. C. Emley, J. C. Sankey, S. I. Kiselev, D. C. Ralph, and R. A. Buhrman, "Time-domain measurements of nanomagnet dynamics driven by spin-transfer torques," *Science*, vol. 307, no. 5707, pp. 228–231, Jan. 2005.
- [34] S. I. Kiselev *et al.*, "Current-induced nanomagnet dynamics for magnetic fields perpendicular to the sample plane," *Phys. Rev. Lett.*, vol. 93, no. 3, p. 036601, Jul. 2004.
- [35] Q. Mistral *et al.*, "Current-driven microwave oscillations in current perpendicular-to-plane spin-valve nanopillars," *Appl. Phys. Lett.*, vol. 88, no. 19, p. 192507, 2006.
- [36] B. L. Miller and D. E. Goldberg, "Genetic algorithms, tournament selection, and the effects of noise," *Complex Syst.*, vol. 9, no. 3, pp. 193–212, 1995.
- [37] M. Takahashi and H. Kita, "A crossover operator using independent component analysis for real-coded genetic algorithms," in *Proc. Congr. Evol. Comput.*, May 2001, pp. 643–649.
- [38] K. Deb, A. Pratap, S. Agarwal, and T. Meyarivan, "A fast and elitist multiobjective genetic algorithm: NSGA-II," *IEEE Trans. Evol. Comput.*, vol. 6, no. 2, pp. 182–197, Apr. 2002.
- [39] J.-G. Zhu, "Magnetoresistive random access memory: The path to competitiveness and scalability," *Proc. IEEE*, vol. 96, no. 11, pp. 1786–1798, Nov. 2008.
- [40] K. L. Wang, J. G. Alzate, and P. K. Amiri, "Low-power non-volatile spintronic memory: STT-RAM and beyond," *J. Phys. D, Appl. Phys.*, vol. 46, no. 7, p. 074003, 2013.
- [41] R. Sbiaa, H. Meng, and S. N. Piramanayagam, "Materials with perpendicular magnetic anisotropy for magnetic random access memory," *Phys. Status Solidi (RRL)*, vol. 5, no. 12, pp. 413–419, 2011.
- [42] T. M. Maffitt *et al.*, "Design considerations for MRAM," *IBM J. Res. Develop.*, vol. 50, no. 1, pp. 25–39, Jan. 2006.
- [43] W. S. Zhao, T. Devolder, Y. Lakys, J. O. Klein, C. Chappert, and P. Mazoyer, "Design considerations and strategies for high-reliable STT-MRAM," *Microelectron. Rel.*, vol. 51, nos. 9–11, pp. 1454–1458, 2011.
- [44] R. Dittrich, T. Schrefl, D. Suess, W. Scholz, H. Forster, and J. Fidler, "A path method for finding energy barriers and minimum energy paths in complex micromagnetic systems," *J. Magn. Magn. Mater.*, vol. 250, pp. 12–19, Sep. 2002.
- [45] R. Dittrich, T. Schrefl, H. Forster, D. Suess, W. Scholz, and J. Fidler, "Energy barriers in magnetic random access memory elements," *IEEE Trans. Magn.*, vol. 39, no. 5, pp. 2839–2841, Sep. 2003.